Quarterly Review of Economics and Finance 104 (2025) 102042

Contents lists available at ScienceDirect

Quarterly Review of Economics and Finance

journal homepage: www.elsevier.com/locate/qref

From algorithms to invention: AI's impact on corporate innovation output and efficiency

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ARTICLE INFO

JEL classification:
O31
O32
L25
M11
D22

Keywords:
Artificial intelligence
Corporate innovation
Corporate strategy
Data assetization
Supply-chain diversification

ABSTRACT

This paper explores how artificial intelligence (AI) drives corporate innovation. Using a procurement-based AI adoption index, patent records, and supply-chain data from 4004 A-share firms over 2011–2023, we find that greater AI adoption significantly increases the output and efficiency of firms' innovation. We propose a dual-channel model in which AI enhances knowledge creation and reuse (knowledge orchestration) and transforms data into actionable environmental assets (data assetization). Heterogeneity analysis reveals that large incumbents and growth-stage firms leverage AI most effectively for innovation outputs and efficiency. Further analysis shows that AI-driven innovation is amplified in firms with executives who have information technology backgrounds, and that highly innovative firms diversify their supply chain to reduce resource risk. Our results demonstrate AI's potential to advance corporate innovation. We conclude with policy recommendations for municipal planners and corporate strategists to enhance firms' competitive advantage and promote the development of AI.

1. Introduction

Artificial intelligence (AI) has become a central driver of corporate innovation, reshaping how firms create new products, processes, and business models (Johnson et al., 2022; Liu et al., 2024a, 2024b). Facing intense competition and sustainability pressures, companies use AI not just for automation but to transform innovation from incremental improvement into a dynamic, systemic process (Ayman Wael Al-Khatib, 2023). This paper examines artificial intelligence's role in corporate innovation. We argue that AI's impact extends through enhanced efficiency, quality, and strategic repositioning across value chains.

A burgeoning body of literature has documented the transformative effects of AI on innovation paradigms. Bahoo et al. (2023) offer a comprehensive taxonomy that situates AI as a general-purpose technology capable of redefining corporate research and development (R&D) processes. Dong et al. (2025) and Zhong and Song (2025) show that AI boosts both the quantity and quality of innovation, mediated by human capital, management, and financing constraints. Wang, Wang, et al. (2024) link digital technology to process innovation, while Guo and Xu (2024) analyze urban digital intelligence transformation, underscoring AI's wider societal and economic impact. Jiang et al. (2024) examine how AI reconfigures corporate innovation, and Kim et al. (2022) highlight its role in driving social innovation.

Despite these insights, important gaps remain. First, many studies adopt a unidimensional perspective by focusing predominantly on either innovation output or efficiency, thereby neglecting the dual-channel effects that AI may exert across multiple innovation dimensions. Second, conventional measurement proxies—often reliant on patent counts or self-reported AI adoption metrics—fail to capture the nuanced interplay between AI-enabled technological integration and complementary organizational assets such as data governance and executive technical expertise. Third, while prior work explores firm size and

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https://doi.org/10.1016/j.qref.2025.102042
Received 6 May 2025; Received in revised form 31 August 2025; Accepted 1 September 2025
Available online 3 September 2025

lifecycle heterogeneity (e.g., Hussain et al., 2024), few examine their interaction with supply chain diversification and ecosystem reconfiguration (Rikap, 2024; Song et al., 2024).

Addressing these lacunae, this study makes several critical contributions. First, it introduces a refined, procurement-based measure of AI adoption that mitigates the limitations of traditional proxies by leveraging comprehensive tender data. Second, it delineates a dual-channel framework that explicates how AI simultaneously amplifies innovation output and enhances innovation efficiency, mediated by mechanisms of knowledge orchestration and data assetization. Third, through rigorous heterogeneity analyses across firm size and lifecycle stages, the study elucidates the contingent effects of AI on innovation, thereby reconciling dynamic capability and resource-based perspectives. Finally, by extending the inquiry to consider the implications for supply chain diversification and strategic leadership, this research offers a holistic account of how AI reconfigures both internal innovation processes and external competitive dynamics.

2. Theoretical analysis and research hypotheses

2.1. AI as a general-purpose technology in innovation

The theoretical foundation of this study rests on the concept of artificial intelligence as a general-purpose technology capable of pervasive innovation-enhancing effects (Andergassen et al., 2017). AI's adaptability, scalability, and complementarity suggest its potential to boost both the volume and productivity of corporate innovation. While traditional inputs such as R&D and human capital face diminishing returns, AI's automation and predictive analytics create nonlinear productivity gains. This theorization predicts that AI adoption should exhibit positive elasticity on patent output (H1a) and innovation efficiency (H1b), as firms leverage machine learning to automate knowledge recombination and optimize resource allocation (Brea, 2024). Furthermore, the general-purpose technology lens anticipates disproportionate returns for organizations with preexisting complementary assets (e.g., data infrastructure, technical talent), laying groundwork for examining heterogeneous effects across firm characteristics.

2.2. Mediating mechanisms: knowledge orchestration and data assetization

Dynamic capability theory provides the scaffolding for hypothesizing AI's innovation mechanisms. The ability to "sense, seize, and transform" knowledge resources (Teece, 2018) is amplified by AI's dual capacity for exploration and exploitation. Firms deploying AI tools can simultaneously identify novel technological trajectories (knowledge creation) and refine existing knowledge repositories (knowledge reuse), suggesting balanced mediation effects between these pathways (H2a). Concurrently, AI implementation necessitates structural changes in data management practices (Al Halbusi et al., 2025). The data assetization mechanism posits that even marginal improvements in data governance—catalyzed by AI adoption—generate recursive innovation returns through enhanced data quality and algorithmic performance (H2b). This theorized mechanism resolves the paradox of why data-rich firms without AI underperform, as raw data stocks only translate to innovation advantages when coupled with AI-driven systematization.

2.3. Heterogeneity in innovation returns: scale and lifecycle dynamics

Resource-based view (RBV) and lifecycle theories jointly inform hypotheses about contingent AI impacts (Chang, 2022; Ma & Wu, 2024). Large firms' superior innovation returns (H3a) stem from three RBV postulates: (1) VRIN (Valuable, Rare, Imperfectly Imitable, Non-substitutable) resources in proprietary datasets and cloud infrastructure, (2) absorptive capacity to integrate AI with legacy systems, and (3) slack resources for iterative experimentation. Conversely, small firms' constrained complementary assets limit their ability to convert AI adoption into measurable innovation gains (Chattopadhyay et al., 2024). Lifecycle theory further predicts that growth-phase firms (H3b) will outperform mature and declining peers in leveraging AI for innovation, as strategic flexibility and growth orientation enable ambidextrous innovation strategies. In contrast, structural inertia and short-termism in declining firms neutralize AI's potential, even when adoption levels match those of growth-phase counterparts.

2.4. Executive expertise as an innovation amplifier

Upper echelons theory and absorptive capacity frameworks converge to hypothesize the moderating role of IT-proficient leadership (H4). Information technology expertise at the executive level enhances AI's innovation impact through two channels: (1) strategic prioritization of AI projects with high knowledge recombination potential, and (2) organizational translation of technical capabilities into business value. This effect is asymmetric: it strengthens patent output (H4a) more than efficiency (H4b), as technically fluent leaders often favor exploratory R&D over process optimization (Bevilacqua et al., 2025).

2.5. Quality-valorization nexus in AI-driven innovation

The studies of innovation quality emphasize that AI's impact extends beyond patent quantity to influence technological significance (H5). Machine learning algorithms, by detecting latent connections across knowledge domains and simulating combinatorial possibilities, enable inventions with higher originality and impact (Martin et al., 2023). This quality enhancement is theorized to operate independently of leadership traits, emerging instead from systemic organizational learning and AI's inherent capacity for pattern recognition in large-scale knowledge corpora. The stability of quality effects across citation metrics (including non-self-citations) further suggests that AI's quality premium reflects genuine technological breakthroughs rather than strategic gaming of innovation metrics (Huang et al., 2025).

2.6. Innovation spillovers in the supply chain

The theoretical implications of AI-driven innovation extend beyond organizational boundaries to reshape firms' positions within industrial ecosystems. Building on resource dependence theory and innovation ecosystem frameworks (Başak Canboy & Wafa Khlif, 2025), this study hypothesizes that technological breakthroughs enabled by AI alter power dynamics in supply chains (H6). High innovation output firms, through patent-driven technological diversification, gain bargaining power to reduce supplier concentration (H6a) by developing alternative inputs or vertical integration capabilities. Conversely, innovation efficiency improvements empower firms to reconfigure customer relationships (H6b) through two mechanisms: (1) cost-optimized customization that attracts diverse client segments, and (2) predictive analytics enabling proactive market expansion.

Quantity and efficiency serve distinct functions: patent-intensive innovation disrupts upstream dependence, while efficiency-driven innovation strengthens downstream ties. Suppliers face higher switching costs than customers, creating asymmetry in ecosystem effects (MacKay, 2022). High-efficiency innovators also achieve multiplier effects: they set technical standards and secure key partners while retaining flexibility. This aligns with Baldwin, 2008 "thin crossing points" view—AI-enabled firms become indispensable nodes in networks, commanding resources and market access through technology rather than contracts.

Fig. 1 shows the conceptual framework of AI and enterprise innovation.

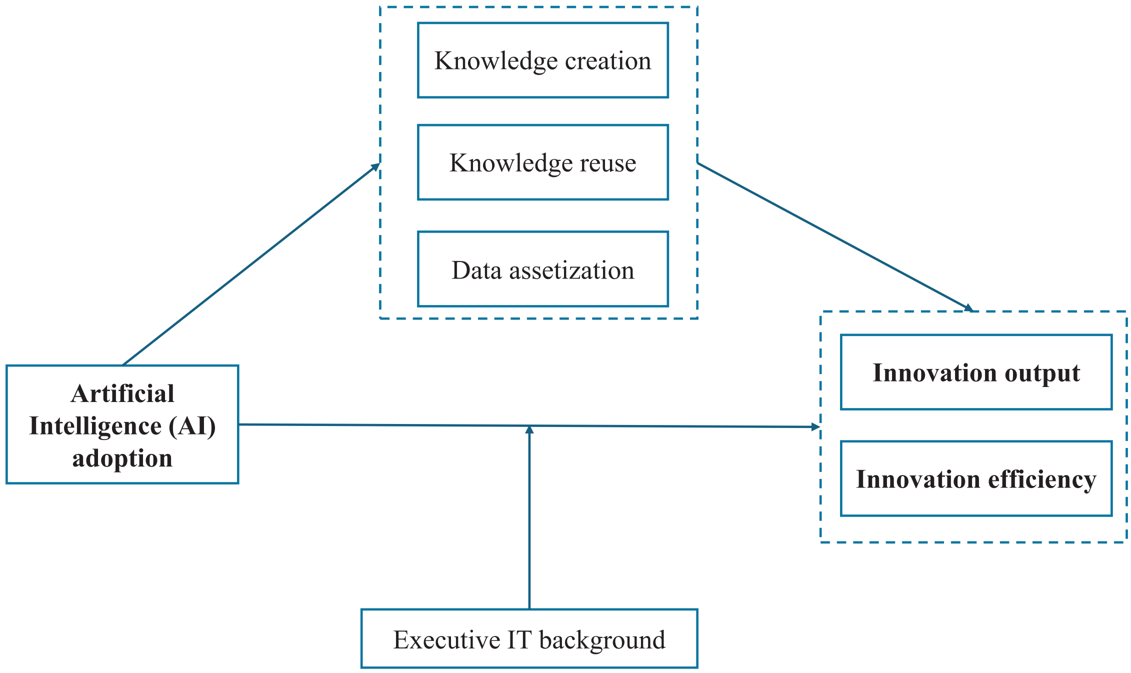

Fig. 1. Conceptual framework of artificial intelligence and enterprise innovation.

3. Research design

3.1. Model specification

To empirically examine the relationship between AI adoption and corporate innovation, we employ a two-way fixed effects model to mitigate omitted variable bias. The baseline specifications are formulated as follows:

$$Ln(1+inno\_output_{i,t}) = \alpha_0 + \alpha_1 Ln(1+AI\_adoption_{i,t}) + \gamma X_{i,t} + \eta_i + \mu_t + \varepsilon_{i,t} \tag{1}$$

$$Inno\_eff_{i,t} = \beta_0 + \beta_1 Ln(1+AI\_adoption_{i,t}) + \gamma X_{i,t} + \eta_i + \mu_t + \varepsilon_{i,t} \tag{2}$$

where i denotes the firm, and t represents the year. $Ln(1+inno\_output_{i,t})$ refers to the innovation output of firm i in year t and $Inno\_eff_{i,t}$ refers to the innovation efficiency of firm i in year t. $Ln(1+AI\_adoption_{i,t})$ indicates the AI adoption level of firm i in year t, quantified through the firm's bidding activities related to AI-related projects. The vector $X_{i,t}$ comprises a set of control variables that account for potential confounding factors influencing corporate innovation. η_i and μ_t represent firm-specific fixed effects and year-specific fixed effects, respectively, which control for other unobserved confounders that do not vary across firms or over time. Finally, $\varepsilon_{i,t}$ is the random error term. To address potential estimation biases arising from residuals and heteroscedasticity, we cluster all standard errors at the firm level.

3.2. Variable definitions

This section delineates the operationalization of variables employed in our empirical analysis, encompassing dependent variables, the focal independent variable, control variables, and mechanism variables. Table 1 presents the variable definitions.

3.2.1. Dependent variables

Corporate innovation is captured through two dimensions: innovation output and innovation efficiency. First, innovation output is measured as the natural logarithm of one plus the number of invention patent applications filed by a firm in a given year ($Ln(1+inno\_output_{i,t})$). This logarithmic transformation mitigates skewness inherent in raw patent counts while preserving the interpretability of innovation scale (Han et al., 2020). Second, innovation efficiency ($Inno\_eff_{i,t}$) is calculated as the ratio of ln(1 +patent applications) to ln(1 +R&D expenditures), reflecting the productivity of R&D investments. This metric addresses concerns about diminishing returns to innovation inputs by quantifying how effectively firms convert R&D spending into patentable outputs.

3.2.2. Core independent variable

The key explanatory variable, AI adoption ($Ln(1+AI\_adoption_{i,t})$), quantifies a firm's engagement in AI-related activities. To construct this variable, we leverage the SSIRD database, which contains over 1.47 million tendering events from Chinese listed firms. Following Buarque et al. (2020) and World Intellectual Property Report (2019), we built an AI terminology dictionary through literature review and expert validation, including terms (For details, please refer to AI-related keyword dictionary in Appendix) like "machine learning," "deep learning," "computer vision," and "neural network." Each tender's project title was scanned against this dictionary. Events matching any keyword were classified as AI-related (coded as 1) and aggregated at the firm-year level. The natural logarithm of the count plus one is then applied to normalize the distribution, yielding a continuous measure of AI adoption intensity. From the initial dataset, 44,600 AI-related tenders were identified, providing granular insights into firms' strategic AI investments (Gans, 2024).

3.2.3. Control variables

To isolate the causal effect of AI adoption on innovation, we include a battery of controls spanning corporate governance, financial structure, and ownership characteristics (Beisland et al., 2014; Cheng, 2025; Chindasombatcharoen et al., 2021; Chu & Hoang, 2020; Harasheh et al., 2022; Wang, Qi, et al., 2024). Corporate governance variables include board size (*lnBoard*), the proportion of independent directors (*lnIndep*), and a binary indicator for CEO-chair duality (*Dual*). Financial structure is proxied by leverage (*Lev*), firm size (*lnSize*), and return on assets (*ROA*). Ownership structure controls encompass executive stock incentives (lnMshare), ownership concentration (*lnTop3*), and state

Table 1
Variable definitions.

Variable	Definition
Explained variable	
Ln (1 + inno_output)	Firm Innovation Output: We measure it by log(1 + the number of invention patent applications)
Inno_eff	Firm Innovation Efficiency: We measure it by log(1 + the number of invention patent applications)/log(1 + R&D expenditures)
Interest variable	
Ln (1 + AI_adoption)	Firm's AI Adoption Level: This is assessed using data from the SSIRD database, which includes over 1.47 million tendering events of publicly listed companies. The AI-related terminology dictionary is constructed from an extensive review of relevant literature and expert judgment, incorporating terms such as AI, automation, machine translation, machine learning, computer vision, deep learning, neural networks, data mining, feature recognition, speech synthesis, speech recognition, large models, augmented reality, multimodal systems, visual models, virtual avatars, simulation training, visual detection, and visual systems. The "project title" of each tender is analyzed using the dictionary, and if any of the terms appear, the event is classified as related to AI (coded as "1"); otherwise, it is coded as "0". From over 1.47 million data samples, 44,600 AI-related tender events are identified.
Control variables	
lnBoard	Board Size: We calculate it by log(the number of board members).
lnIndep	Proportion of Independent Directors: We calculate it by log (the percentage of independent directors on the board).
Lev	Leverage Ratio: We use the total liabilities to total assets ratio as a measure of leverage.
lnSize	Firm size: we calculate it by log (total assets at the fiscal year-end).
lnAge	Firm age, we calculate it by log (the number of years since the firm's establishment).
lnMshare	Executive Stock Ownership Incentives: we calculate it by log (the ratio of the number of shares held by management to the total shares of the firm).
lnTop3	Ownership concentration, we gauge it using log(the proportion of shares held by the top three shareholders).
lnTobinQ	Firm value, we measure it by log(the firm's market value to / (total assets - net intangible assets - net goodwill)).
ROA	Return on Assets: This is calculated as net profit divided by the average total assets.
Dual	Dual role integration, where the positions of general manager and chairman are held by the same individual is coded as 1, otherwise 0.
SOE	Ownership Type: State-owned enterprises are coded as 1, and others as 0.
Mechanism variables	
lnknow_crea	Knowledge Creation: We define knowledge creation patents as those whose IPC classification combinations have never appeared before within a five-year period. These patents are aggregated on a firm-year basis, and the natural logarithm of the sum is taken to obtain this variable.
lnkonw_reuse	Knowledge Reuse: Knowledge reuse patents are those whose IPC classification combinations have appeared previously within a five-year period. These patents are aggregated on a firm-year basis, and the natural logarithm of the sum is taken to obtain this variable.
lnda	Data Assetization: We measure this various through text analysis. Using relevant legal and regulatory texts related to data assets as the corpus, and employing seed words such as "information," "network," "digital," and "data," we apply the Word2Vec neural network model and deep learning techniques to construct a set of similar words. The frequency of these similar words is counted, and the natural logarithm of the frequency plus one is taken to quantify this various.
da_people	Investment in data management personnel. This variable is measured as the ratio of the number of online job postings related to data management to the total number of online job postings, aggregated at the firm-year level for listed companies.

ownership (*SOE*). Additionally, market valuation (*lnTobinQ*) and firm age (*lnAge*) are incorporated to account for lifecycle and investor expectations. These variables collectively address potential confounders such as resource availability, governance quality, and market pressures that might influence innovation outcomes independently of AI adoption.

3.2.4. Mechanism variables

Three mechanism variables are introduced to explore the channels through which AI adoption affects innovation. First, knowledge creation (*lnknow_crea*) is operationalized as patents with novel International Patent Classification (IPC) combinations that have not appeared in the preceding five years. This aligns with the knowledge-based view (KBV), which posits that innovation arises from the recombination of previously unconnected knowledge elements. Second, knowledge reuse (*lnknow_reuse*) captures patents leveraging existing IPC combinations to solve new problems, reflecting KBV's emphasis on refining and recombining established knowledge. Third, data assetization (*lnda*) quantifies firms' capabilities in converting data into strategic assets. Employing seed words such as "information," "network," "digital," and "data," we apply the Word2Vec neural network model and deep learning techniques to construct a set of similar words, with the natural logarithm of the frequency plus one to measure data utilization intensity. This variable aligns with theories of data-driven innovation, which emphasize how data assetization enables novel business models and operational efficiencies.

3.3. Data sources and descriptive statistics

The empirical analysis draws on a comprehensive panel dataset spanning 2011–2023, covering 4004 publicly listed firms across 266 Chinese cities, resulting in 27,978 firm-year observations. The data integration process involved multiple sources to ensure robustness and granularity.

Primary data on AI adoption originates from the SSIRD (State Statistical Information Resource Database) tender platform, which archives over 1.47 million procurement events from listed firms. To identify AI-related projects, we developed a keyword dictionary through iterative reviews of AI literature and consultations with domain experts, ensuring alignment with evolving AI terminologies. Project titles were scanned for matches with terms such as "neural networks" and "machine learning," and binary indicators were aggregated annually at the firm level. Patent data for innovation output and knowledge creation/reuse variables were extracted from the China National Intellectual Property Administration (CNIPA) database, which provides detailed invention patent applications classified by International Patent Classification (IPC) codes. Financial and governance variables, including R&D expenditures, leverage, and board characteristics, were also sourced from the China Stock Market & Accounting Research (CSMAR), a widely cited repository for corporate financial data in China. Textual analysis for data assetization employed annual reports from listed firms, processed using a Word2Vec model trained on a corpus of Chinese regulatory documents and data-related legislation. Seed words "information" "network" "digital" "data" were expanded into a synonym set through neural network embeddings, and term frequencies were log-transformed to construct the final measure. Ownership type (SOE) and executive roles (Dual) were verified against official stock exchange disclosures to ensure accuracy.

Table 2 summarizes the distribution of key variables. The dependent variable innovation output (*Ln(1 +inno_output)*) exhibits a mean of 2.133 (SD = 1.458), with a median of 2.079, indicating moderate right-skewness. Innovation efficiency (*Inno_eff*) averages 0.164 (SD = 0.080), suggesting substantial heterogeneity in R&D productivity across firms. The core independent variable AI adoption (*Ln(1 +AI_adoption)*) shows limited engagement overall (mean = 0.050, SD = 0.278), though maximum values (2.079) highlight early adopters driving upper-tail variation. Control variables reflect typical corporate profiles in

Table 2
Descriptive statistics.

VarName	Obs	Mean	SD	Min	Median	Max
Ln(1+ inno_output)	27,978	2.133	1.458	0.000	2.079	6.182
Inno_eff	27,978	0.164	0.080	0.000	0.174	0.332
Ln(1+ AI_adoption)	27,978	0.050	0.278	0.000	0.000	2.079
lnknow_crea	27,978	1.378	1.270	0.000	1.386	7.712
lnkonw_reuse	27,978	0.800	1.173	0.000	0.000	8.875
lnda	27,978	3.361	0.730	0.000	3.258	6.979
da_people	27,978	0.040	0.080	0.000	0.000	1.000
lnBoard	27,978	2.101	0.191	1.609	2.197	2.565
lnIndep	27,978	0.319	0.037	0.288	0.310	0.452
Lev	27,978	0.388	0.195	0.049	0.378	0.864
lnSize	27,978	22.090	1.212	20.008	21.900	25.957
lnAge	27,978	2.935	0.323	1.946	2.996	3.584
lnMshare	27,978	−3.705	3.453	−13.122	−2.115	−0.354
lnTop3	27,978	−0.795	0.335	−1.772	−0.753	−0.197
lnTobinQ	27,978	0.586	0.450	−0.150	0.500	2.021
ROA	27,978	0.039	0.064	−0.251	0.042	0.199
Dual	27,978	0.355	0.478	0.000	0.000	1.000
SOE	27,978	0.216	0.411	0.000	0.000	1.000

emerging markets. Firm size (*lnSize*) averages 22.090 (SD = 1.212), consistent with the dominance of large enterprises in China's listed sector. Leverage (*Lev*) stands at 0.388 (SD = 0.195), aligning with moderate debt levels observed in manufacturing-heavy economies. Ownership concentration (*lnTop3*) and executive stock incentives (*lnMshare*) show negative skewed distributions, reflecting concentrated ownership and limited equity participation among management. Mechanism variables reveal distinct innovation pathways. Knowledge creation (*lnknow_crea*) has a mean of 1.378 (SD = 1.270), while knowledge reuse (*lnknow_reuse*) averages 0.800 (SD = 1.173), underscoring firms' greater reliance on incremental innovation. Data assetization (*lnda*) displays higher baseline activity (mean = 3.361, SD = 0.730), consistent with China's rapid digital transformation. The sample's temporal (13-year span) and spatial (266 cities) diversity enhances representativeness, capturing regional disparities and policy shifts in AI and innovation. Notably, 21.6 % of observations represent state-owned enterprises (*SOE*), and 35.5 % exhibit CEO-chair duality (*Dual*), mirroring governance trends in China's corporate landscape.

4. Main results

Before proceeding to causal inference, we conduct a correlation analysis to assess preliminary relationships between variables and potential multicollinearity risks. Table 3 reports pairwise Pearson correlation coefficients. First, the focal independent variable, AI adoption (*Ln (1 +AI_adoption)*), exhibits statistically significant positive correlations with firm size (*lnSize*, ρ = 0.203), age (*lnAge*, ρ = 0.138), and state ownership (SOE, ρ = 0.201). This aligns with theoretical expectations: larger, older, and state-owned firms in China may possess greater resources and institutional support to implement AI initiatives. Conversely, AI adoption shows negative correlations with executive stock incentives (*lnMshare*, ρ = −0.155) and CEO-chair duality (*Dual*, ρ = −0.072), suggesting that firms with concentrated managerial ownership or leadership separation are less likely to prioritize AI investments. Second, control variables demonstrate expected interdependencies. Firm size (*lnSize*) is strongly correlated with leverage (*Lev*, ρ = 0.533), reflecting larger firms' access to debt financing. Board size (*lnBoard*) and independent director ratio (*lnIndep*) are negatively correlated (ρ = −0.582), suggesting larger boards dilute independent directors' influence. Ownership concentration (*lnTop3*) shows a weak positive correlation with profitability (*ROA*, ρ = 0.233), consistent with principal-agent theory that dominant shareholders drive performance. Notably, the correlation matrix reveals no severe multicollinearity concerns. Most pairwise coefficients remain below 0.5, the threshold

Table 3
Correlation test.

	Ln(1+AI_adoption)	*lnBoard*	*lnIndep*	*Lev*	*lnSize*	*lnAge*	*lnMshare*	*lnTobinQ*	*lnTop3*	*ROA*	*Dual*	*SOE*
Ln(1+ AI_adoption)	1											
lnBoard	0.075***	1										
lnIndep	0.007	−0.582***	1									
Lev	0.079***	0.152***	−0.028***	1								
lnSize	0.203***	0.252***	−0.031***	0.533***	1							
lnAge	0.138***	0.075***	−0.021***	0.193***	0.270***	1						
lnMshare	−0.155***	−0.238***	0.065***	−0.346***	−0.455***	−0.278***	1					
lnTobinQ	−0.078***	−0.114***	0.040***	−0.298***	−0.378***	−0.084***	0.140***	1				
lnTop3	0.025***	−0.050***	0.059***	−0.105***	0.002	−0.148***	0.017***	−0.059***	1			
ROA	0.011*	0.008	−0.003	−0.365***	−0.009	−0.100***	0.128***	0.189***	0.233***	1		
Dual	−0.072***	−0.178***	0.124***	−0.137***	−0.189***	−0.113***	0.255***	0.093***	0.033***	0.035***	1	
SOE	0.201***	0.277***	−0.083***	0.279***	0.369***	0.212***	−0.630***	−0.151***	0.016***	−0.075***	−0.277***	1

commonly used to flag multicollinearity. The strongest correlation exists between firm size and leverage (ρ = 0.533). These preliminary results underscore the importance of controlling for firm characteristics and governance mechanisms when isolating AI's impact on innovation. The absence of extreme correlations between AI adoption and other variables mitigates concerns about confounding biases dominating the baseline results.

4.1. Baseline regression results

The baseline regression results, presented in Table 4, reveal a statistically and economically significant relationship between AI adoption and corporate innovation performance. Columns (1) and (3) estimate the effect on innovation output (*Ln(1 +inno_output)*), while columns (2) and (4) focus on innovation efficiency (*Inno_eff*). Models (3) and (4) include all controls and fixed effects, yielding the most robust estimates. A 1 % increase in AI adoption intensity raises patent applications by 0.145 % (β = 0.145, $p < 0.01$) after accounting for firm characteristics and fixed effects. This elasticity suggests that AI adoption meaningfully enhances firms' capacity to generate patented inventions, likely by automating R&D processes, enabling large-scale data analysis, and reducing trial-and-error costs.

Table 4
Baseline regression results.

	(1) *Ln (1+inno_output)*	(2) *Inno_eff*	(3) *Ln (1+inno_output)*	(4) *Inno_eff*
Ln (1 + AI_adoption)	0.135***	0.005***	0.145***	0.005***
	(0.028)	(0.002)	(0.027)	(0.002)
lnBoard			0.133	0.007
			(0.087)	(0.005)
Dual			0.018	0.001
			(0.024)	(0.001)
lnIndep			0.632*	0.028
			(0.365)	(0.022)
Lev			−0.253***	−0.011**
			(0.090)	(0.005)
lnSize			0.539***	0.023***
			(0.029)	(0.002)
lnAge			0.111	0.016
			(0.207)	(0.012)
lnMshare			0.013**	0.001**
			(0.006)	(0.000)
lnTobinQ			0.099***	0.003**
			(0.027)	(0.002)
lnTop3			0.050	0.004
			(0.063)	(0.004)
ROA			−0.067	−0.000
			(0.140)	(0.008)
SOE			0.025	0.002
			(0.063)	(0.003)
_cons	2.128***	0.164***	−10.476***	−0.412***
	(0.001)	(0.000)	(0.857)	(0.047)
Firm_fe	yes	yes	yes	yes
Year_fe	yes	yes	yes	yes
N	27,852	27,852	27,852	27,852
adj. R^2	0.698	0.658	0.714	0.668

Notes: This table examines the impact of artificial intelligence (AI) adoption on firms innovation, including both the output and efficiency of innovation. The variable *Ln(1+ AI_adoption)* serves as the primary explanatory variable, representing the extent of AI-related projects in a firm's bidding activities, thereby reflecting the firm's level of AI integration. *Ln(1+ inno_output)* indicates the innovation output, measured by the natural logarithm of (1+the number of invention patents application). *Inno_eff* represents innovation efficiency, calculated as the ratio of the natural logarithm of (1+the number of invention patent application) to the natural logarithm of (1+R&D expenditure). Control variables remain consistent with the baseline regression. Individual and year fixed effects are controlled for. Standard errors are clustered at the individual level to address heteroscedasticity, with robust standard errors reported in parentheses. Statistical significance is indicated at the 1 %, 5 %, and 10 % levels by ***, **, and *, respectively.

For innovation efficiency, where the dependent variable remains in its original ratio form, the coefficient (β = 0.005, $p < 0.01$) indicates that a 1 % increase in AI adoption raises innovation efficiency by 0.005 units. This represents a substantial 3.05% (0.005/0.164) increase relative to the sample mean innovation efficiency level. This aligns with AI's dual role in both augmenting patent output and optimizing R&D resource allocation—for instance, through predictive analytics for project selection or AI-driven simulations that reduce experimental waste.

Control variables follow patterns consistent with innovation literature. Firm size (*lnSize*) positively correlates with innovation output (β = 0.539, $p < 0.01$) and efficiency (β = 0.023, $p < 0.01$), reflecting R&D scale economies. Leverage (*Lev*) negatively affects innovation (β = −0.253, $p < 0.01$), consistent with debt constraints limiting risky R&D. Executive stock incentives (*lnMshare*) positively influence innovation (β = 0.013, $p < 0.05$), supporting agency theory that equity aligns managers with long-term goals.

These findings support the hypothesis that AI adoption acts as a general-purpose technology, reshaping both the volume and productivity of corporate innovation. The results are consistent with theoretical frameworks emphasizing AI's role in reducing knowledge recombination costs and accelerating iterative problem-solving.

Our findings align with and extend the emerging body of research on AI's role in corporate innovation. The positive elasticity of AI adoption on patent output (β = 0.145) resonates with Jiang et al. (2024), who identified a increase in patent applications per AI adoption using Chinese listed firms. Stronger estimates may reflect temporal advancements: the 2023 data captures faster AI deployment in R&D than the 2011–2022 panel. Our innovation efficiency results (β = 0.005) support Zhong and Song (2025), showing AI adoption boosts innovation by improving resource allocation, though eco-innovation may yield larger gains. Firm characteristics moderate AI's impact. Like Hussain et al. (2024), who found private firms benefit more from AI, our *SOE* coefficient (β = 0.025, $p > 0.1$) suggests state-owned firms gain less—possibly due to bureaucratic inertia. This contrasts with Dong et al. (2025), who reported SOEs as main AI beneficiaries in innovation, highlighting that AI's effects vary with policy context and innovation type.

4.2. Robustness checks

To validate the reliability of our baseline findings, we conduct a battery of robustness tests to address some concerns on variable measurement, model specification, sensitivity analysis, and causal identification.

4.2.1. Alternative AI adoption measures

To verify the robustness of our baseline AI adoption measure, we introduce three alternative indicators. First, we apply an inverse hyperbolic sine (IHS) transformation to the AI adoption variable to address the issue of zero inflation (*AI_ihs*). This transformation retains the advantages of a logarithmic specification while accommodating zero values, making it suitable for datasets with a substantial number of zeros. Second, we construct a text-based AI intensity measure *Ln (1 + AI_adoption1)* from firms' annual reports, which captures the rhetorical emphasis placed on AI-related topics in corporate disclosures and offers a complementary perspective beyond quantitative adoption metrics. Third, we use a two-period lagged AI adoption variable to account for potential delayed effects of AI implementation on innovation outcomes and to mitigate concerns about reverse causality.

As shown in Appendix Table A1, the estimated coefficients remain positive and statistically significant across all specifications, indicating that our core findings are highly robust. Importantly, the economic magnitudes of the effects are broadly consistent across models. The IHS-transformed specification gives a slightly smaller coefficient than the baseline, but the direction and meaning remain unchanged, indicating the AI–innovation link is not driven by functional form. Similarly, the lagged specification produces a coefficient of comparable size to the contemporaneous model, underscoring the persistence of the AI effect. These results reinforce the credibility of our conclusions.

4.2.2. High-dimensional fixed effects

We further incorporate high-dimensional fixed effects—specifically, city-year and industry-city fixed effects—to address potential confounding factors arising from heterogeneity at the local and industry levels (see Appendix Table A2). The city-year fixed effects control for unobserved shocks, policy changes, or economic conditions that vary over time within cities and could otherwise bias the estimated relationship between AI adoption and innovation. The industry-city fixed effects account for time-invariant factors specific, such as localized industry clusters, historical technological bases, or persistent institutional differences. Including these fixed effects helps isolate firm-level variation in AI adoption that is independent of broader regional or industry-level trends. The results remain largely consistent, with only minor variations, which are expected given the more stringent controls. This pattern supports the view that, while regional and industry trends contribute to differences in innovation, firm-level AI adoption continues to exert an independent and economically meaningful effect.

4.2.3. Changing clustering approaches

Moreover, we alter the clustering approach by employing two-way clustering at the city-year level (see Appendix Table A3). This method accounts for potential correlations in the error terms among firms operating within the same city and year, which may arise from shared exposure to local economic shocks, policy environments, or unobserved city-year-specific factors. Although this clustering approach results in slightly wider confidence intervals, the core estimates remain statistically significant, and the economic magnitude of the coefficients is largely preserved. This underscores that our conclusions are not sensitive to the method used for calculating standard errors.

4.2.4. Alleviating confounding factors

We further address potential confounding factors by controlling for a set of policy-related variables (see Appendix Table A4). Specifically, we include provincial patent subsidies (*patent_sub*), municipal intellectual property institutions (*know_por*), and municipal smart city pilot programs (*smart_city*) in our regressions. These controls help alleviate concerns that our results are driven by local innovation policies rather than firm-level AI adoption. The inclusion of these innovation policy does not materially alter the estimated effect of AI adoption. The main coefficients remain largely unchanged or show a slight decline, but the overall pattern and substantive interpretation remain consistent.

4.2.5. Sensitivity test for the AI-related projects

To assess the robustness of our AI adoption index, we construct two alternative keyword dictionaries and re-estimated the main regression models accordingly. First, we exclude a set of semantically ambiguous keywords that may lead to misclassification, such as "unmanned," "algorithm," and "recognition." Second, we introduce a more refined set of AI-related terms by disaggregating generic expressions like "intelligent" into more specific applications, including "intelligent regulation," "intelligent advisory," "intelligent voice," and "intelligent customer service."[1]; The estimation results using these alternative dictionaries, reported in Appendix Table A5, Columns (1)–(4), remain highly consistent with those of the baseline model, indicating that our findings are robust to reasonable variations in keyword construction.

In addition, to address the potential misclassification of non-AI-related projects arising from the incidental presence of AI-related terms in project titles, we conduct a manual review of all 44,600 procurement projects containing AI keywords. We identify a small number of cases where the keywords appeared in the names of listed companies (e.g., containing the term "intelligent") but are not substantively related to AI applications. After excluding these projects, we re-estimate our models (Appendix Table A5, Columns 5–6). The results remain stable and significant, confirming the accuracy of our AI adoption index and the robustness of our findings.

4.2.6. Alleviating selective bias

To further address potential selection bias in our estimation, we employ two complementary approaches: propensity score matching (PSM) and the Heckman two-stage model.

First, the PSM method allows us to construct a matched sample of firms with similar observable characteristics, thereby mitigating bias that may arise if AI adopters systematically differ from non-adopters in ways that affect innovation outcomes. We estimate the propensity scores—the predicted probabilities of AI adoption—using firm-level covariates, including all control variables at the firm level. Using 1:1 nearest neighbor matching, AI-adopting firms are matched with non-adopting firms exhibiting similar propensity scores, ensuring a more balanced comparison group. The effect of AI adoption on innovation outcomes is then re-estimated using this matched sample.

In Appendix Table A6, we find that the estimated coefficients for AI adoption become larger after matching compared to the baseline regression (e.g., the coefficient for innovation output increases from 0.145 to 0.206, and for innovation efficiency increases from 0.005 to 0.007). This increase suggests that the innovation-enhancing effect of AI adoption becomes more pronounced once firms are matched on key observable characteristics. The results indicate that in the unmatched sample, some non–AI-adopting firms may systematically differ in ways that attenuate the observed relationship. The matched sample offers a cleaner counterfactual and reinforces the economic relevance of AI adoption for innovation. Furthermore, Appendix Figure A1 shows a substantial reduction in covariate imbalance after matching, confirming the reliability of the matching procedure.

Second, we use the Heckman two-stage model to predict firm innovation based on whether senior executives have overseas experience, which affects innovation likelihood but not its scale or efficiency.

In the first stage, we estimate a probit model where the dependent variable is the firm's probability of engaging in innovation activity, proxied by two alternative innovation indicators (*inno_dummy1* and *inno_dummy2*). From this model, we calculate Inverse Mills Ratio (*IMR1* and *IMR2*) to capture potential selection effects.

In the second stage, we incorporate the corresponding Inverse Mills Ratio into the baseline regression of AI adoption on innovation outcomes. This procedure helps alleviate bias arising from non-random firm behavior that may jointly determine both AI adoption and innovation performance. The results (reported in Appendix Table A7) demonstrate that our main findings remain robust after accounting for sample selection bias, further reinforcing the validity of our conclusions.

4.3. Instrumental variable method

To address potential endogeneity concerns in corporate AI adoption—such as reverse causality or omitted variable bias—this paper employs an instrumental variable (IV) approach to estimate the model. Specifically, we use the industry average level of AI adoption (excluding the firm itself) as the instrument to capture the exogenous variation in firm-level AI adoption (see Table 5). The validity of this IV is discussed based on two underlying assumptions.

The IV must be strongly correlated with the endogenous explanatory variable. In our study, this IV is expected to significantly influence a firm's own AI adoption. This is because firms are typically subject to competitive pressure, imitation effects, and technological spillovers within their industry. When AI adoption is prevalent in an industry,

[1] The specific keywords can be found in Alternative AI-related keyword dictionary in the Appendix.

Table 5
Instrumental variable method.

	(1)	(2)	(3)	(4)
	Ln(1+ AI_adoption)	*Ln(1+ inno_output)*	*Ln(1+ AI_adoption)*	*Inno_eff*
IV	0.644***		0.644***	
	(0.102)		(0.102)	
Ln (1 + AI_adoption)		0.679***		0.022**
		(0.202)		(0.011)
control	yes	yes	yes	yes
Firm_fe	yes	yes	yes	yes
Year_fe	yes	yes	yes	yes
Industry_fe	yes	yes	yes	yes
N	27,745	27,745	27,745	27,745
first-stage F-statistic	39.51		39.51	
Kleibergen-Paap rk LM statistic	29.638***		29.638***	

Notes: This table adopts the instrumental variable method to alleviate the endogeneity problem. The variable *Ln(1+ AI_adoption)* serves as the primary explanatory variable, representing the extent of AI-related projects in a firm's bidding activities, thereby reflecting the firm's level of AI integration. *Ln(1+ inno_output)* indicates the innovation output, measured by the natural logarithm of (1+the number of invention patents application). *Inno_eff* represents innovation efficiency, calculated as the ratio of the natural logarithm of (1 +the number of invention patent application) to the natural logarithm of (1 + R&D expenditure). *IV* represents the average annual adoption level of artificial intelligence at the industry level (excluding itself), and industries are classified based on two-digit industry codes. Control variables remain consistent with the baseline regression model. We control for firm, year and industry fixed effects. Standard errors are clustered at the individual level to address heteroscedasticity, with robust standard errors reported in parentheses. Statistical significance is indicated by ***, **, and * for the 1 %, 5 %, and 10 % levels, respectively.

firms are more motivated to adopt AI technologies to maintain competitive advantage or avoid falling behind. Our first-stage regression results confirm this relationship: the IV has a significant positive effect on firm-level AI adoption (first-stage coefficient = 0.644, $p < 0.01$). Moreover, the first-stage F-statistic is well above the conventional threshold of 10 ($F = 39.51$), effectively ruling out concerns about weak instruments.

The IV should affect innovation output only through its impact on corporate AI adoption, and not have a direct effect on firm innovation. Industry-level technology adoption reflects broader industry trends and the external environment, but it does not directly intervene in the innovation decisions of individual firms. By including industry fixed effects, we further account for potential confounding influences at the industry level on firm innovation. Therefore, the exclusion restriction is theoretically justified.

The IV estimation results indicate that, after addressing endogeneity concerns, the positive effect of corporate AI adoption on both innovation output and innovation efficiency becomes stronger compared to the potentially downward-biased TWFE estimate. This suggests that failing to account for endogeneity may lead TWFE to underestimate the true contribution of AI adoption to firm innovation. Our findings provide robust evidence that AI adoption meaningfully enhances both innovation output and efficiency.

5. Mechanism analysis

To elucidate the pathways through which artificial intelligence enhances corporate innovation, we empirically test three theoretically grounded mechanisms derived from the knowledge-based view of the firm (Grant, 1996) and AI-enabled innovation frameworks (Sahoo et al., 2024; Rikap, 2024). Our analysis focuses on knowledge creation, knowledge reuse, and data assetization — three interrelated channels through which AI is hypothesized to reshape innovation processes. We estimate a series of mechanism models to disentangle AI's indirect innovation effects:

$$lnknow\_crea_{i,t} = \delta_0 + \delta_1 Ln(1 + AI\_adoption_{i,t}) + \gamma X_{i,t} + \eta_i + \mu_t + \varepsilon_{i,t} \tag{3}$$

$$lnknow\_reuse_{i,t} = \delta_0 + \delta_1 Ln(1 + AI\_adoption_{i,t}) + \gamma X_{i,t} + \eta_i + \mu_t + \varepsilon_{i,t} \tag{4}$$

$$lnda_{i,t} = \delta_0 + \delta_1 Ln(1 + AI\_adoption_{i,t}) + \gamma X_{i,t} + \eta_i + \mu_t + \varepsilon_{i,t} \tag{5}$$

where $lnknow\_crea_{i,t}$ is defined as the level of knowledge creation of firm i in year t. $lnknow\_reuse_{i,t}$ represents the level of knowledge reuse for firm i in year t. $lnda_{i,t}$ refers to the data assetization level of firm i in year t. The remaining variables remain consistent with those in the baseline regression.

The selection of these mechanisms is motivated by AI's dual role as a combinatorial innovation engine (Weitzman, 1998) and organizational capability enhancer (Teece, 2018). First, machine learning algorithms excel at identifying non-obvious knowledge linkages across domains (Brynjolfsson & Mcafee, 2017), which should amplify both creation and reuse of technological components. Second, AI implementation necessitates systematized data pipelines (Cockburn et al., 2019), transforming raw information into codified assets that reduce R&D transaction costs.

Table 6 demonstrates statistically and economically significant effects across all examined channels. The estimated elasticities for knowledge creation (0.247) and knowledge reuse (0.234) suggest that a 1 % increase in AI adoption is associated with approximately a 0.25 % increase in the generation of new knowledge and a 0.23 % increase in the recombination of existing knowledge. The closeness of these elasticities indicates that AI adoption plays a comparable role in fostering both radical (knowledge creation) and incremental (knowledge reuse) innovation activities.

This dual impact can be attributed to the inherent characteristics of AI technologies. On one hand, AI enables firms to process unstructured data, identify novel patterns, and generate original solutions, thereby catalyzing breakthrough innovation. On the other hand, AI also

Table 6
The mechanism test of artificial intelligence to enterprise innovation.

	(1) *lnknow_crea*	(2) *lnkonw_reuse*	(3) *lnda*	(4) *da_people*
Ln(1+ AI_adoption)	0.247***	0.234***	0.025**	0.007***
	(0.027)	(0.028)	(0.012)	(0.002)
control	yes	yes	yes	yes
Firm_fe	yes	yes	yes	yes
Year_fe	yes	yes	yes	yes
N	27,852	27,852	27,852	27,852
adj. R^2	0.676	0.725	0.804	0.326

Notes: This report examines the mechanism test of artificial intelligence to corporate innovation. The variable *Ln(1+ AI_adoption)* serves as the primary explanatory variable, representing the extent of AI-related projects in a firm's bidding activities, thereby reflecting the firm's level of AI integration. *lnknow_crea* measures corporate knowledge creation, defined as the logarithm of the number of patents in the past five years that represent unique combinations of IPC (International Patent Classification) codes never previously observed. *lnkonw_reuse* quantifies corporate knowledge reuse, represented as the logarithm of the number of patents in the past five years that reuse specific combinations of IPC codes previously recorded. And the *lnda* indicates the level of corporate data assetization, calculated by applying Python-based text analysis to annual reports of listed companies to determine the frequency of terms related to "data assets," with the logarithmic transformation of the resulting count. The *da_people* measures the ratio of the number of personnel recruitment positions related to data management posted online by an enterprise to all the online recruitment positions posted by the enterprise. Control variables remain consistent with the baseline regression. We control for both individual and year fixed effects. Standard errors are clustered at the individual level to address heteroscedasticity, with robust standard errors reported in parentheses. Statistical significance is indicated at the 1 %, 5 %, and 10 % levels by ***, **, and *, respectively.

enhances firms' capacity to search, retrieve, and recombine existing knowledge elements efficiently, thereby supporting incremental innovation. This interpretation aligns with Zhang et al. (2025), who provide empirical evidence that generative AI tools empower employees to engage in both breakthrough creativity and incremental improvement. Although our study focuses on broader AI adoption at the firm level, the underlying mechanisms—pattern recognition, automation of cognitive tasks, and augmentation of human capabilities—are consistent. Therefore, our findings highlight the distinctive capability of AI to serve as a general-purpose technology that enables innovation along both breakthrough and incremental trajectories. This sheds light on why AI functions not merely as a tool of efficiency, but also as a catalyst for creative knowledge transformation across the innovation spectrum.

The data assetization channel yields a modest yet statistically significant coefficient under both the baseline and alternative specifications. Our estimates show that AI adoption increased by 1%, and the data assetization increased by 0.025% (effect = 0.025 %, SE = 0.012, $p < 0.05$). Data assetization, in turn, sustains innovation through feedback loops between data quality and algorithm performance. To address concerns about measurement validity, we introduce an alternative proxy—namely, the share of data management–related recruitment (*da_people*). We collected online job postings from listed companies and calculated the firm-year proportion of positions in data management—such as data governance, architecture, and security—relative to total postings. This alternative indicator captures actual organizational resource allocation to data-related capabilities and serves as a meaningful proxy for institutionalized data infrastructure development. We find that *da_people* produces a consistent and more precisely estimated coefficient of 0.007 % (SE = 0.002, $p < 0.01$). These results suggest that AI's infrastructural demands promote institutionalized data governance. While the magnitudes are small in absolute terms, they reinforce recent evidence that incremental improvements in data systematization can yield outsized returns in AI-enabled R&D scalability (Li et al., 2025).

Our results support dynamic capability theory (Teece, 2018): AI's value derives not from data stockpiles but from firms' ability to orchestrate knowledge flows. This resolves the paradox observed by Zhong and Song (2025), where data-rich firms underperformed in innovation unless paired with AI analytics.

The symmetry between knowledge creation ($\beta = 0.247$) and reuse ($\beta = 0.234$) contrasts with sector-specific studies where reuse dominated sustainability innovation (Zhong & Song, 2025). This discrepancy suggests AI's knowledge effects are context-contingent: general innovation may require balanced exploration-exploitation, while mission-driven innovation (e.g., decarbonization) prioritizes targeted reuse of eco-technologies. Practically, the findings underscore that firms deploying AI for innovation must invest equally in combinatorial search algorithms and knowledge graph technologies to maximize both novelty and efficiency gains.

6. Heterogeneity analysis

6.1. Firm size as a contingency factor

Our results reveal stark disparities in AI's innovation returns across firm sizes (Table 7). For large firms (Columns 2 and 4), AI adoption significantly enhances both innovation output ($\beta = 0.154$, $p < 0.01$) and efficiency ($\beta = 0.006$, $p < 0.01$). In contrast, small firms exhibit economically and statistically negligible effects (output=0.069, $p > 0.1$; efficiency=0.001, $p > 0.1$). The 122 % larger coefficient for innovation output in large firms ($\Delta\beta = 0.085$) aligns with Babina et al. (2024) finding that AI-driven growth concentrates among industry leaders, amplifying market dominance. This size premium likely stems from three interrelated advantages: (1) complementary asset stocks (e.g., proprietary data reserves and cloud infrastructure), (2) organizational readiness to integrate AI with legacy systems, and (3) risk tolerance for iterative AI experimentation. Our results extend their framework by demonstrating that these scale advantages manifest most strongly in innovation outcomes rather than operational metrics like sales growth.

Table 7
Artificial intelligence and firm innovation: the role of firm size.

	Ln (1 + inno_output)		*Inno_eff*	
	(1) *Size_small*	(2) *Size_big*	(3) *Size_small*	(4) *Size_big*
Ln(1+AI_adoption)	0.069	0.154***	0.001	0.006***
	(0.064)	(0.031)	(0.003)	(0.002)
control	yes	yes	yes	yes
Firm_fe	yes	yes	yes	yes
Year_fe	yes	yes	yes	yes
N	15,553	12,015	15,553	12,015
adj. R^2	0.618	0.759	0.618	0.706

Notes: This table investigates the differential impact of firm size on the adoption of Artificial Intelligence and its subsequent empowerment of corporate innovation. The variable *Ln (1 + AI_adoption)* serves as the primary explanatory variable, representing the extent of AI-related projects in a firm's bidding activities, thereby reflecting the firm's level of AI integration. *Ln(1+ Patent)* indicates the innovation output, measured by the natural logarithm of (1+the number of invention patents application). *Inno_eff* represents innovation efficiency, calculated as the ratio of the natural logarithm of (1 +the number of invention patent application) to the natural logarithm of (1 + R&D expenditure). We categorize firms based on the average size, with odd-numbered columns representing samples from smaller firms and even-numbered columns representing samples from larger firms. Control variables remain consistent with the baseline regression. We control for both individual and year fixed effects. Standard errors are clustered at the individual level to address heteroscedasticity, with robust standard errors reported in parentheses. Statistical significance is indicated at the 1 %, 5 %, and 10 % levels by ***, **, and *, respectively.

The null effects for small firms challenge narratives about AI democratizing innovation access. While cloud-based AI tools theoretically lower entry barriers, realizing innovation benefits requires complementary investments in data governance and talent development — resources disproportionately available to large firms. Notably, the output gap ($\Delta\beta$ output=0.085) exceeds the innovation efficiency gap ($\Delta\beta$ Inno_eff=0.005), implying large firms achieve disproportionately higher R&D productivity gains — likely through AI-enabled optimization of existing knowledge repositories, as evidenced by their stronger performance in knowledge reuse (Table 7).

6.2. Lifecycle dynamics

Table 8 shows that AI's innovation effects differ across corporate lifecycle stages. Growth-stage firms respond most strongly (βoutput=0.164, $p < 0.01$; βefficiency=0.008, $p < 0.01$), mature firms less so (βoutput=0.116, $p < 0.01$; βefficiency=0.004, $p < 0.05$), and declining firms show no significant effects (βoutput=0.046, $p > 0.1$; βefficiency=0.003, $p > 0.1$). This gradient, based on dividend payouts, revenue growth, capital expenditures, and firm age (Ditillo, 1998), reflects differences in strategic flexibility and resource allocation.

Growth-phase firms leverage AI's potential through ambidextrous innovation strategies, combining exploratory R&D (novel patent combinations) with rapid commercialization. For instance, AI-powered simulation tools allow young tech firms to prototype at scale, compressing development cycles (Nambisan et al., 2019). Mature firms, while benefiting from AI's efficiency gains (βefficiency=0.004), face institutional inertia in reallocating resources from core businesses to AI-driven opportunities. Declining firms' inability to capitalize on AI aligns with the rigidity trap hypothesis (Leonard-Barton, 2019): legacy processes and short-term survival focus impede investments in AI's long-term innovation payoffs.

These findings partially reconcile Babina et al. (2024) observation of AI-driven industry concentration with lifecycle theory. While their study emphasizes size as the primary differentiator, our results suggest

Table 8
Artificial intelligence and firm innovation: the role of enterprise life cycle.

	Ln (1+inno_output)			*Inno_eff*		
	(1)	(2)	(3)	(4)	(5)	(6)
	Growth	*Maturity*	*Decline*	*Growth*	*Maturity*	*Decline*
Ln (1 + AI_adoption)	0.164***	0.116***	0.046	0.008***	0.004**	0.003
	(0.052)	(0.032)	(0.124)	(0.003)	(0.002)	(0.008)
control	yes	yes	yes	yes	yes	yes
Firm_fe	yes	yes	yes	yes	yes	yes
Year_fe	yes	yes	yes	yes	yes	yes
N	11,315	12,487	823	11,315	12,487	823
adj. R^2	0.705	0.745	0.754	0.675	0.687	0.698

Notes: This table investigates the differential impact of corporate life cycle stages. The variable *Ln(1+ AI_adoption)* serves as the primary explanatory variable, representing the extent of AI-related projects in a firm's bidding activities, thereby reflecting the firm's level of AI integration. *Ln(1+Inno_output)* indicates the innovation output, measured by the natural logarithm of (1+the number of invention patents application). *Inno_eff* represents innovation efficiency, calculated as the ratio of the natural logarithm of (1 +the number of invention patent application) to the natural logarithm of (1+ R&D expenditure). We categorize firms into different lifecycle stages using a multivariate ranking approach, considering four factors: dividend payout ratio, growth rate of core business revenue, capital expenditure ratio, and the firm's establishment date. Based on these dimensions, the firm lifecycle is divided into three stages: *growth*, *maturity*, and *decline*. Control variables remain consistent with the baseline regression. We control for both individual and year fixed effects. Standard errors are clustered at the individual level to address heteroscedasticity, with robust standard errors reported in parentheses. Statistical significance is indicated at the 1 %, 5 %, and 10 % levels by ***, **, and *, respectively.

lifecycle stage interacts with size: even large firms in decline derive limited innovation benefits (βoutput=0.046 vs. 0.154 for large firms overall), implying that organizational vitality mediates AI's effectiveness. a 1% rise in AI adoption is associated with a 0.164% increase in firms' patent output — comparable to the annual innovation growth rate of top-performing startups. This underscores AI's role as an amplifier of existing growth trajectories rather than a turnaround solution for struggling firms.

Our size and lifecycle analyses refine Babina et al. (2024) conclusion about AI exacerbating industry concentration. While they attribute dominance to scale per se, our results highlight that scale combined with growth momentum drives AI's innovation premium. The 0.164–0.116–0.046 coefficient gradient across lifecycle stages (Table 8) suggests that AI disproportionately rewards firms with both (1) resource abundance (a size effect) and (2) strategic agility (a lifecycle effect). This dual contingency explains why even some large incumbents fail to harness AI effectively — a pattern observed in traditional manufacturing sectors where mature firms prioritize AI for cost reduction rather than innovation.

Methodologically, our lifecycle measurement approach — synthesizing financial, operational, and temporal indicators (Ditillo, 1998) — addresses limitations in prior studies relying solely on firm age. The results caution against one-size-fits-all AI policy frameworks: growth-stage firms may require subsidies for experimental AI projects, while mature firms need incentives to reorient AI from process automation to innovation generation.

7. Further analysis

7.1. Moderating role of executive IT expertise

To formally assess how executive IT expertise conditions AI's innovation effects, we estimate the following fixed-effects models:

$$Ln(1+inno\_output_{i,t}) = \alpha_0 + \alpha_1 Ln(1+AI\_adoption_{i,t}) + \alpha_2 IT\_high_{i,t} + \alpha_3 Ln(1+AI\_adoption_{i,t}) \times IT\_high_{i,t} + \gamma X_{i,t} + \eta_i + \mu_t + \varepsilon_{i,t} \tag{6}$$

$$Inno\_eff_{i,t} = \beta_0 + \beta_1 Ln(1+AI\_adoption_{i,t}) + \beta_2 IT\_high_{i,t} + \beta_3 Ln(1+AI\_adoption_{i,t}) \times IT\_high_{i,t} + \gamma X_{i,t} + \eta_i + \mu_t + \varepsilon_{i,t} \tag{7}$$

where $IT\_high_{i,t}$ indicates the IT background of senior management. It is measured by whether the proportion of senior executives with an IT background exceeds the mean value; if it does, the value is 1, otherwise it is 0. The IT background includes both educational qualifications and professional experience. The IT educational background refers to degrees in fields such as Electronic Information, Computer Science, E-Commerce, Information and Computing Science, Information Management and Information Systems, and Information Resource Management. Professional experience refers to work experience in areas such as Information Technology, Information Management, Information Systems, Digital Transformation, ERP Systems, Software Development, Internet/Network Development, Computer Engineering, System Engineering, E-Commerce, E-Government, the Internet of Things, and Cloud Computing.

The presence of senior executives with IT backgrounds significantly amplifies AI's impact on innovation, though effects vary across performance dimensions (Table 9). For firms with IT-savvy leaders (IT_high =

Table 9
Discussion on the Executive information technology background.

	(1) *Ln (1 + inno_output)*	(2) *Inno_eff*
Ln(1+ AI_adoption)	0.102***	0.004*
	(0.034)	(0.002)
IT_high	0.240	0.017*
	(0.164)	(0.009)
*Ln(1+ AI_adoption)** IT_high	0.773**	0.016
	(0.329)	(0.018)
control	yes	yes
Firm_fe	yes	yes
Year_fe	yes	yes
N	24,402	24,402
adj. R^2	0.729	0.682

Notes: This table presents the moderating effect of executives' information technology (IT) background on the relationship between artificial intelligence (AI) and firm innovation. The variable *Ln (1 + AI_adoption)* serves as the primary explanatory variable, representing the extent of AI-related projects in a firm's bidding activities, thereby reflecting the firm's level of AI integration. *Ln (1 +Inno_output)* indicates the innovation output, measured by the natural logarithm of (1 +the number of invention patents application). *Inno_eff* represents innovation efficiency, calculated as the ratio of the natural logarithm of (1 +the number of invention patent application) to the natural logarithm of (1 + R&D expenditure). *IT_high* indicates the IT background of senior management. It is measured by whether the proportion of senior executives with an IT background exceeds the mean value; if it does, the value is 1, otherwise it is 0. Control variables remain consistent with the baseline regression. We control for both individual and year fixed effects. Standard errors are clustered at the individual level to address heteroscedasticity, with robust standard errors reported in parentheses. Statistical significance is indicated at the 1 %, 5 %, and 10 % levels by ***, **, and *, respectively.

1), AI adoption interacts positively with IT_high ($\beta = 0.773$, $p < 0.05$). In contrast, the moderating effect on innovation efficiency ($\beta = 0.016$, $p > 0.1$) lacks statistical significance, suggesting IT expertise primarily enhances AI's quantity rather than efficiency of innovation.

These results align with the absorptive capacity framework (Cohen & Levinthal, 1990), where executives' technical competence enables firms to better identify, assimilate, and exploit AI-driven opportunities. The significant interaction effect indicates that, relative to their peers, firms with a higher proportion of executives with IT backgrounds can magnify the innovation output driven by AI. This amplification effect likely operates through two channels: (1) Strategic alignment: IT-experienced executives prioritize AI projects with higher innovation spillovers, avoiding "shiny object syndrome" in technology adoption. (2) Organizational learning: Technically fluent leaders bridge communication gaps between data scientists and business units, accelerating knowledge recombination (Nambisan et al., 2019).

The null result for innovation efficiency ($\beta = 0.016$, $p > 0.1$) reveals a paradox: while IT leadership boosts patent volume, it does not translate to proportional R&D productivity gains. This discrepancy may stem from exploration-exploitation trade-offs (March, 1991). IT-oriented executives might favor ambitious, long-horizon AI projects (e.g., generative models for drug discovery) that inflate R&D costs before yielding efficiency dividends. Alternatively, the lack of efficiency gains could reflect measurement limitations — patent counts capture innovation output but not the economic value or cost-saving potential of AI-driven inventions.

7.2. Influence on innovation quality

The adoption of artificial intelligence not only enhances the quantity but also elevates the quality of corporate innovation, as evidenced by the statistically significant coefficients in Table 10. A 1% increase in AI adoption corresponds to a 0.011 % rise in total patent citations ($\beta = 0.011$, $p < 0.05$) and a 0.010 % increase in citations excluding self-references ($\beta = 0.010$, $p < 0.05$). These effects, while modest in magnitude, demonstrate AI's capacity to amplify the impact and originality of inventions — dimensions critically tied to economic value. The stability across citation metrics (with and without self-citations) suggests that AI's quality-enhancing effects reflect genuine technological breakthroughs rather than strategic citation gaming.

This aligns with evidence that AI aids in identifying novel combinatorial opportunities across distant knowledge domains (Cockburn et al., 2019). For instance, machine learning algorithms can map latent connections between biomedical patents and materials science literature, fostering cross-disciplinary breakthroughs.

Table 10
Discussion on the enterprise innovation quality.

	(1) *lninno_quality*	(2) *lninno_quality1*
Ln (1 + AI_adoption)	0.011**	0.010**
	(0.005)	(0.005)
control	yes	yes
Firm_fe	yes	yes
Year_fe	yes	yes
N	24,461	24,461
adj. R^2	0.648	0.606

Notes: This table reveals the impact of corporate artificial intelligence adoption on innovation quality. The variable *Ln (1 + AI_adoption)* serves as the primary explanatory variable, representing the extent of AI-related projects in a firm's bidding activities, thereby reflecting the firm's level of AI integration. Innovation quality is measured using two distinct proxies: the total number of patent citations (*lninno_quality*) and the number of citations excluding self-citations (*lninno_quality1*), with one added to each count before taking the natural logarithm. Control variables remain consistent with the baseline regression. We control for both individual and year fixed effects. Standard errors are clustered at the individual level to address heteroscedasticity, with robust standard errors reported in parentheses. Statistical significance is indicated at the 1 %, 5 %, and 10 % levels by ***, **, and *, respectively.

Notably, the quality effects operate independently of executive IT expertise — contrasting with Table 10's moderation pattern — which implies that AI's capacity to improve invention significance stems from systemic organizational learning rather than leadership traits alone.

Methodologically, the use of non-self-citations (*lninno_quality1*) mitigates concerns about citation inflation, reinforcing the validity of quality gains. These results refine Babina et al. (2024) focus on product innovation volume by showing AI's equally critical role in innovation valorization. The positive influence of a 0.011% quality premium per 1% AI adoption suggests that firms leveraging AI for innovation could accumulate substantial competitive advantages over time — AI adoption increase would translate to the citation premium within five years, compounding annual R&D productivity.

7.3. Economic and management impact: supply chain diversification

The regression models formalize the hypothesis that AI-driven innovation reshapes corporate supply chain structures by reducing dependence on concentrated partners. It is modeled as functions of innovation performance:

$$Supply\_power_{i,t}(Custom\_power_{i,t}) = \alpha_0 + \alpha_1 Inno\_output\_high_{i,t} + \gamma X_{i,t} + \eta_i + \mu_t + \varepsilon_{i,t} \tag{8}$$

$$Supply\_power_{i,t}(Custom\_power_{i,t}) = \beta_0 + \beta_1 InnoEff\_high_{i,t} + \gamma X_{i,t} + \eta_i + \mu_t + \varepsilon_{i,t} \tag{9}$$

In this model, we focus on the impact of improvements in corporate innovation output and innovation efficiency on the concentration of upstream and downstream supply chains. $Supply\_power_{i,t}$ represents the supplier concentration of firm *i* in year *t*, measured by the proportion of procurement from the firm's top five suppliers. $Custom\_power_{i,t}$ indicates the customer concentration of firm *i* in year *t*, quantified by the sales revenue from the firm's top five customers. $Inno\_output\_high_{i,t}$ is a dummy variable that indicates whether the innovation output of the firm exceeds the mean value. Similarly, $InnoEff\_high_{i,t}$ is a dummy variable representing whether the firm's innovation efficiency is above the mean. All other variables remain consistent with the baseline regression model.

Table 11 shows that AI-driven innovation significantly lowers supply chain concentration, reducing operational risks. Firms with high innovation output (*Inno_output_high* = 1) cut supplier concentration by 0.685 unit ($\beta = -0.685$, $p < 0.01$). Relative to the sample mean value of 33.736, this represents a decrease of approximately 2.03%(0.685/33.736) in procurement reliance on top-five suppliers for an average firm, substantially lowering exposure to supply disruptions (Barrot & Sauvagnat, 2014). For customer concentration, higher innovation efficiency reduces dependence by 0.383 unit ($\beta = -0.383$, $p < 0.1$), while high innovation output has limited effect ($\beta = -0.233$, $p > 0.1$), suggesting that productive R&D helps firms expand market channels.

The asymmetrical impacts between upstream and downstream relationships reveal distinct mechanisms: (1) Supplier diversification primarily benefits from innovation volume (output), as patent-intensive firms develop alternative materials/components through technological breakthroughs, reducing lock-in to dominant suppliers. (2) Customer diversification requires innovation efficiency, where higher R&D productivity allows firms to customize solutions for niche markets while maintaining economies of scale.

The coefficient for *InnoEff_high* on supplier concentration ($\beta = -0.719$, $p < 0.01$) implies a multiplier effect: Compared with the sample average, efficient innovators achieve both greater supplier independence (-2.31%(-0.719/33.736) procurement concentration) and moderate customer base expansion (-1.18%(−0.383/32.370) sales

Table 11
The benefits of enterprise innovation: supply chain diversification.

	(1) *Supply_power*	(2) *Custom_power*	(3) *Supply_power*	(4) *Custom_power*
Inno_output_high	−0.685***	−0.233		
	(0.230)	(0.230)		
InnoEff_high			−0.719***	−0.383*
			(0.216)	(0.222)
control	yes	yes	yes	yes
Firm_fe	yes	yes	yes	yes
Year_fe	yes	yes	yes	yes
N	27,280	27,551	27,280	27,551
adj. R^2	0.741	0.815	0.741	0.815

Notes: This table reveals the impact of corporate innovation on supply chain concentration. *Inno_output_high* indicates a higher level of corporate innovation, where a value of 1 is assigned when the innovation output exceeds the mean, and 0 otherwise. *InnoEff_high* reflects a higher level of innovation efficiency, with a value of 1 when the innovation efficiency surpasses the mean, and 0 otherwise. *Supply_power* denotes the bargaining power of suppliers, measured by the proportion of the procurement amount from the top five suppliers. *Custom_power* represents the bargaining power of customers, gauged by the proportion of sales generated from the top five customers. the mean value of Supply_power is 33.736 and the mean value of Custom_power is 32.370. Control variables remain consistent with the baseline regression. We control for both individual and year fixed effects. Standard errors are clustered at the individual level to address heteroscedasticity, with robust standard errors reported in parentheses. Statistical significance is indicated at the 1 %, 5 %, and 10 % levels by ***, **, and *, respectively.

concentration). This aligns with case evidence from AI-driven manufacturers who leverage predictive analytics to onboard regional suppliers without sacrificing quality.

Economically, highly innovative firms (*Inno_output_high* or *InnoEff_high*=1) exhibit a reduction in supply chain concentration of 0.383-0.719 unit compared with low-innovation firms (*Inno_output_high* or *InnoEff_high*=0). Relative to the mean concentration level , this corresponds to a 1.18%-2.31% decrease. Although the coefficient appears modest, the presence of bullwhip effects in supply chains suggests that even small reductions in concentration can be amplified, implying a substantially larger economic and operational impact than the raw percentage might indicate. The adjusted $R2$ values (0.741–0.815) confirm that innovation metrics explain substantial variation in supply chain configurations beyond conventional determinants like firm size or industry.

8. Conclusions

This study provides evidence that artificial intelligence drives corporate innovation by expanding scale and improving efficiency. By combining detailed AI adoption metrics with supply chain and patent data, we show that AI reshapes firms' strategies in knowledge creation, risk management, and market positioning, beyond mere productivity gains. AI boosts patent output and improves innovation efficiency —effects larger than those of traditional digital technologies. The benefits vary across contexts: large, expanding firms capture more gains, whereas IT-savvy leaders enhance patent output more than others, owing to superior absorptive capacity. AI acts as a capability amplifier, with value emerging from its fit with data practices, lifecycle stage, and leadership skills.

The results advance innovation economics theory by resolving the tension between dynamic capabilities and resource-based views. AI's dual-channel impact—simultaneously enabling knowledge exploration and exploitation —demonstrates that technological plasticity, not resource stocks, determines firms' ability to harness disruptive innovations. This plasticity is institutionalized through the underappreciated data assetization pathway, where marginal AI investments trigger exponential returns via recursive data-AI quality loops. Our lifecycle-stage analysis further contextualizes the AI dominance paradox: growth-phase firms achieve annual patent output gains through ambidextrous innovation strategies, while declining firms remain trapped in rigidity cycles despite equivalent AI adoption levels. These findings collectively reposition AI from a productivity tool to an architectural innovation that reconfigures organizational DNA.

Practitioners must navigate three strategic imperatives. First, AI investments require precise alignment with innovation objectives—output maximization for supplier diversification versus efficiency optimization for customer base broadening. Second, data governance frameworks must evolve from IT projects to C-suite priorities, as the long-term gains from data assetization accumulate into sustainable competitive advantages. Third, leadership development programs should bridge technical and strategic competencies to resolve the IT expertise paradox, where patent volume surges outpace R&D productivity gains.

For policymakers, uniform AI promotion is insufficient; targeted strategies are needed. The absence of significant effects among small and medium-sized enterprises (SMEs) and firms experiencing a declining cycle highlights the need for tailored approaches that address their unique barriers to AI adoption. For instance, municipal planners could design public procurement tenders that explicitly incentivize SMEs to collaborate with AI vendors or participate in AI-oriented consortia. Such mechanisms can facilitate knowledge transfer and help bridge the adoption gap among smaller firms. Furthermore, tiered subsidy schemes could be structured to link financial support not only to metrics of innovation efficiency but also to demonstrable efforts by SMEs to integrate AI through collaborative initiatives. This approach would better align public funding with substantive adoption practices. In addition, local governments could establish AI service centers or open-source platforms that provide affordable access to technical expertise, tools, and best practices—thereby democratizing AI adoption across firms with varying levels of absorptive capacity. Finally, industrial policy could more proactively leverage AI-driven innovation as a means of enhancing supply chain resilience. This recommendation is grounded in our finding that supplier concentration elasticity among efficient innovators reflects AI's potential to mitigate systemic supply risks, particularly amid escalating geopolitical uncertainties.

This study has limitations. Procurement-based AI measures may undercount in-house development, and patent-based quality metrics could be complemented with commercialization data. Still, our findings position AI as a transformative force that rewards agile firms, penalizes rigid ones, and redefines innovation. Success depends on treating AI as a capability that reshapes innovation ecosystems, not just a tool to automate tasks. Future research should examine how generative AI affects the quantity-quality balance and whether blockchain can enhance data assetization. For theory, competitive advantage now comes from continually evolving the innovation process, not merely adopting technology. Firms that thrive will treat AI as a capability to evolve, not an asset to deploy.

Declaration of competing interest

The authors declare that they have no conflicts of interest related to this work.

Acknowledgements

This research was supported by the National Social Science Foundation of China (NO.: 24XJY028) and the Postgraduate Research & Practice Innovation Program of Jiangsu Province (No.: KYCX25_3503).

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.qref.2025.102042.

Data availability

The data used to support the findings of this study are included within the article, and further inquiries can be directed to the corresponding author.

References

Al Halbusi, H., Al-Sulaiti, K. I., Alalwan, A. A., & Al-Busaidi, A. S. (2025). AI capability and Green innovation impact on sustainable performance: Moderating role of big data and knowledge management. *Technological Forecasting and Social Change, 210*, Article 123897. https://doi.org/10.1016/j.techfore.2024.123897

Andergassen, R., Nardini, F., & Ricottilli, M. (2017). Innovation diffusion, general purpose technologies and economic growth. *Structural Change and Economic Dynamics, 40*, 72–80. https://doi.org/10.1016/j.strueco.2016.12.003

Ayman Wael Al-Khatib. (2023). Drivers of generative artificial intelligence to fostering exploitative and exploratory innovation: A TOE framework. *Technology in Society, 75*. https://doi.org/10.1016/j.techsoc.2023.102403, 102403–102403.

Babina, T., Fedyk, A., He, A., & Hodson, J. (2024). Artificial intelligence, firm growth, and product innovation. *Journal of Financial Economics, 151*(1). https://doi.org/10.1016/j.jfineco.2023.103745, 103745–103745.

Bahoo, S., Cucculelli, M., & Qamar, D. (2023). Artificial intelligence and corporate innovation: A review and research agenda. *Technological Forecasting and Social Change, 188*, Article 122264. https://doi.org/10.1016/j.techfore.2022.122264

Baldwin, C. Y. (2008). Where do transactions come from? Modularity, transactions, and the boundaries of firms. *Industrial and corporate change, 17*(1), 155–195.

Barrot, J.-N., & Sauvagnat, J. (2014). Input specificity and the propagation of idiosyncratic shocks in production networks. *SSRN Electronic Journal*. https://doi.org/10.2139/ssrn.2427421

Başak Canboy, & Wafa Khlif. (2025). Beyond efficiency: Revisiting AI platforms, servitization and power relations from a critical perspective. *International Journal of Production Economics*. https://doi.org/10.1016/j.ijpe.2025.109550, 109550–109550.

Beisland, L. A., Mersland, R., & Randøy, T. (2014). The association between microfinance rating scores and corporate governance: a global survey. *International Review of Financial Analysis, 35*, 268–280. https://doi.org/10.1016/j.irfa.2014.10.012

Bevilacqua, S., Masárová, J., Perotti, F. A., & Ferraris, A. (2025). Enhancing top managers' leadership with artificial intelligence: Insights from a systematic literature review. *Review of Managerial Science*. https://doi.org/10.1007/s11846-025-00836-7

Brea, E. (2024). The yin yang of AI: Exploring how commercial and non-commercial orientations shape machine learning innovation. *Research Policy, 53*(6), Article 105008. https://doi.org/10.1016/j.respol.2024.105008

Brynjolfsson, E., & Mcafee, A. (2017). *The Business of Artificial Intelligence How AI Fits Into Your Data Science Team*. 〈https://starlab-alliance.com/wp-content/uploads/2017/09/The-Business-of-Artificial-Intelligence.pdf〉.

Buarque, B. S., Davies, R. B., Hynes, R. M., & Kogler, D. F. (2020). OK computer: The creation and integration of AI in Europe. *Cambridge Journal of Regions, Economy and Society, 13*(1), 175–192. https://doi.org/10.1093/cjres/rsz023

Chang, W. (2022). The effectiveness of AI salesperson vs. Human salesperson across the buyer-seller relationship stages. *Journal of Business Research, 148*, 241–251. https://doi.org/10.1016/j.jbusres.2022.04.065

Chattopadhyay, S., Honoré, F., & Won, S. (2024). Free range startups? Market scope, academic founders, and the role of general knowledge in AI. *Strategic Management Journal*. https://doi.org/10.1002/smj.3685

Cheng, L. (2025). Political governance and firm performance in China: Evidence from a quasi-natural experiment. *Journal of Financial Stability, 76*, Article 101348. https://doi.org/10.1016/j.jfs.2024.101348

Chindasombatcharoen, P., Chatjuthamard, P., Jiraporn, P., & Treepongkaruna, S. (2021). Achieving sustainable development goals through board size and innovation. *Sustainable Development*. https://doi.org/10.1002/sd.2264

Chu, L. K., & Hoang, D. P. (2020). The impact of corporate political activity on innovation in Sub-Saharan African countries: A double-edged sword. *Borsa Istanbul Review, 20*(4), 358–374. https://doi.org/10.1016/j.bir.2020.04.004

Cockburn, I.M., Henderson, R., & Stern, S. (2019). *The Impact of Artificial Intelligence on Innovation*. 115–148. https://doi.org/10.7208/chicago/9780226613475.003.0004.

Cohen, W. M., & Levinthal, D. A. (1990). Absorptive capacity: A new perspective on learning and innovation. *Administrative Science Quarterly, 35*(1), 128–152. https://doi.org/10.2307/2393553

Ditillo, A. (1998). Intellectual Capital - Navigating in the New Business Landscape981Johan Roos, Göran Roos, Leif Edvinsson, Nicola C. Dragonetti. *Intellectual Capital - Navigating in the New Business Landscape*. London: Macmillan Press 1997. 143 pp, ISBN: 0-333-69479-1 £25.00. Business Process Management Journal, 4(1), 85–88. https://doi.org/10.1108/bpmj.1998.4.1.85.1.

Dong, X., Zhou, N., Zhao, X., & Yang, S. (2025). The impact of artificial intelligence on corporate Green innovation: can "increasing quantity" and "improving quality" go hand in hand? *Journal of Environmental Management, 376*. https://doi.org/10.1016/j.jenvman.2025.124439, 124439–124439.

Gans, J. S. (2024). How learning about harms impacts the optimal rate of artificial intelligence adoption. *Economic Policy*. https://doi.org/10.1093/epolic/eiae053

Grant, R. M. (1996). Toward a Knowledge-Based theory of the firm. *Strategic Management Journal, 17*(S2), 109–122. https://doi.org/10.1002/smj.4250171110

Guo, X., & Xu, J. (2024). Can urban digital intelligence transformation promote corporate Green innovation? Evidence from China. *Journal of Environmental Management, 371*, Article 123245. https://doi.org/10.1016/j.jenvman.2024.123245

Han, P., Liu, C., & Tian, X. (2020). Does trading spur specialization? Evidence from patenting. *SSRN Electronic Journal*. https://doi.org/10.2139/ssrn.3681360

Harasheh, M., Capocchi, A., & Amaduzzi, A. (2022). Capital structure in family firms: The role of innovation activity and institutional investors. *EuroMed Journal of Business*. https://doi.org/10.1108/emjb-12-2021-0191

Huang, K. G., Su, Y.-S., Chen, J., & Kajikawa, Y. (2025). Shaping the future through developing and managing breakthrough innovations: A new conceptual framework. *Technological Forecasting and Social Change, 214*. https://doi.org/10.1016/j.techfore.2025.124039, 124039–124039.

Hussain, M., Yang, S., Umer Sahil Maqsood, & Ammar, M. (2024). Tapping into the Green potential: The power of artificial intelligence adoption in corporate Green innovation drive. *Business Strategy and the Environment*. https://doi.org/10.1002/bse.3710

Jiang, L., Xuan, Y., & Zhang, K. (2024). Unlocking innovation potential: The impact of artificial intelligence transformation on enterprise innovation capacity. *European Journal of Innovation Management*. https://doi.org/10.1108/ejim-07-2024-0809

Johnson, P. C., Laurell, C., Ots, M., & Sandström, C. (2022). Digital innovation and the effects of artificial intelligence on firms' research and development – Automation or augmentation, exploration or exploitation? *Technological Forecasting and Social Change, 179*(179), Article 121636. https://doi.org/10.1016/j.techfore.2022.121636

Kim, E., Jang, G.-Y., & Kim, S.-H. (2022). How to apply artificial intelligence for social innovations. *Applied Artificial Intelligence*, 1–22. https://doi.org/10.1080/08839514.2022.2031819

Leonard-Barton, D. (2019). Core capabilities and core rigidities: A paradox in managing new product development. *Strategic Management Journal, 13*(S1), 111–125. https://doi.org/10.1002/smj.4250131009

Li, G., Guo, B., He, T., & Xu, H. (2025). Artificial intelligence and corporate Green development: The role of innovation efficiency. *Applied Economics Letters*, 1–5. https://doi.org/10.1080/13504851.2025.2479078

Liu, P., Lai, Y., & Liu, D. (2024a). Artificial intelligence research in organizations: A bibliometric approach. *Cogent Business Management, 11*(1). https://doi.org/10.1080/23311975.2024.2408439

Liu, X., Cifuentes-Faura, J., Yang, X., & Pan, J. (2024b). The Green innovation effect of industrial robot applications: Evidence from Chinese manufacturing companies. *Technological Forecasting and Social Change, 210*, Article 123904. https://doi.org/10.1016/j.techfore.2024.123904

Ma, D., & Wu, W. (2024). Does artificial intelligence drive technology convergence? Evidence from Chinese manufacturing companies. *Technology in Society, 79*, Article 102715. https://doi.org/10.1016/j.techsoc.2024.102715

MacKay, A. (2022). Contract duration and the costs of market transactions. *American Economic Journal: Microeconomics, 14*(3), 164–212. https://doi.org/10.1257/mic.20200128

March, J. G. (1991). Exploration and exploitation in organizational learning. *Organization Science, 2*(1), 71–87. https://doi.org/10.1287/orsc.2.1.71

Martin, Y., Dai, X., & Deng, Z. (2023). Using machine learning to investigate consumers' emotions: The spillover effect of AI defeating people on consumers' attitudes toward AI companies. *Internet Research*. https://doi.org/10.1108/intr-02-2022-0113

Nambisan, S., Wright, M., & Feldman, M. (2019). The digital transformation of innovation and entrepreneurship: Progress, challenges and key themes. *Research Policy, 48*(8), Article 103773.

Rikap, C. (2024). Varieties of corporate innovation systems and their interplay with global and national systems: Amazon, facebook, google and Microsoft's strategies to produce and appropriate artificial intelligence. *Review of International Political Economy, 31*(6), 1–29. https://doi.org/10.1080/09692290.2024.2365757

Sahoo, S., Kumar, S., Donthu, N., & Singh, A. K. (2024). Artificial intelligence capabilities, open innovation, and business performance – Empirical insights from multinational B2B companies. *Industrial Marketing Management, 117*, 28–41. https://doi.org/10.1016/j.indmarman.2023.12.008

Song, Y., Zhang, Y., Zhang, Z., & Sahut, J.-M. (2024). Artificial intelligence, digital finance, and Green innovation. *Global Finance Journal, 64*, Article 101072. https://doi.org/10.1016/j.gfj.2024.101072

Teece, D. J. (2018). Business models and dynamic capabilities. *Longest Range Planning, 51*(1), 40–49. https://doi.org/10.1016/j.lrp.2017.06.007

Wang, H., Qi, J., Li, Z., Sensoy, Ahmet, & Xing, H. (2024). Excessive financialization and "Original sin Theory": Redemption from corporate reputation. *Research in International Business and Finance, 70*. https://doi.org/10.1016/j.ribaf.2024.102295, 102295–102295.

Wang, J., Wang, X., Sun, F., & Li, X. (2024). The functional mechanisms through which artificial intelligence influences the innovation of Green processes of enterprises. *Systems, 12*(9). https://doi.org/10.3390/systems12090378, 378–378.

Weitzman, M. L. (1998). Recombinant growth. *The Quarterly Journal of Economics, 113*(2), 331–360. https://doi.org/10.1162/003355398555595

Zhang, X., Yu, P., & Ma, L. (2025). How and when generative AI use affects employee incremental and radical creativity: An empirical study in China. *European Journal of Innovation Management*. https://doi.org/10.1108/EJIM-04-2024-0466

Zhong, K., & Song, L. (2025). Artificial intelligence adoption and corporate Green innovation capability. *Finance Research Letters, 72*, Article 106480. https://doi.org/10.1016/j.frl.2024.106480